

Quantifying soil macropore networks in different forest communities using industrial computed tomography in a mountainous area of North China

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Received: 2 December 2015 / Accepted: 8 May 2016 / Published online: 27 May 2016
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Abstract

Purpose Macropores have important effects on the movement of soil water, air, and chemical substances. However, the quantitative relationship between complex 3D soil macropore networks and forest communities remains unclear in the northern mountainous area in China. The objectives of this study were to (1) use industrial computed tomography (CT) scanning and image analysis to quantitatively analyze macropore networks in intact soil columns and (2) identify characteristics of soil macropore networks in different forest communities.

Materials and methods Intact soil columns (100-mm diameter, 300 mm long) were taken from six local forest communities with three replicates for a total of 18 samples. Industrial X-ray CT was used to scan soil samples; then, the scanned images were used to obtain the 3D images of rock fragments and macropore structures. Next, the macropore structure was quantified, including volume, diameter, surface area, length, angle, tortuosity, and number of macropores. This technique provided an accurate method to quantify the structure of macropores.

Results and discussion The analysis and results revealed that different forest communities influence soil macropore 3D structure significantly and in different ways. Macropores in mixed *Pinus tabulaeformis*, *Castanea mollissima*, and *Ulmus pumila* forest had the largest diameter, surface area, network density, and length density of macropores as well as the smallest mean tortuosity of soil macropores. This is caused by the fact that mixed forest soils had more complex root systems, better soil structure, and more biotic activity. Within the soils of a single forest community, macropore porosity, network density, surface area density, and length of macropores decreased with increased soil depth, because more roots and more biological activity were present in the surface soil.

Conclusions Advanced industrial CT technology can allow an accurate quantification of soil macropore structure. This is important because this type of structure has significant effects on soil water, air, and chemical transport. The results suggest that mixed forest is the best afforestation model in the northern mountainous area in China because of its ability to improve soil structure.

Keywords Forest community · Industrial computed tomography · Macropore · Rock fragment

Responsible editor: Zhihong Xu

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1 Introduction

Macropores are created by root channels, earthworm burrows, fissures, and inter-aggregate voids and play a central role in the transport of nutrients, water, air, chemicals, and pollutants in soil (Lin et al. 2005; Katuwal et al. 2015; Jarvis 2007; Lamandé and Schjønning 2011). Soil indices such as soil thickness, density, macropore porosity, water content, hydraulic

parameters, mechanical composition, occurrence layer, and organic matter all affect water movement (Hallett et al. 2009). The influence of macropores on water flow depends mainly on the arrangement and 3D characteristics of macropores, including the length, diameter, volume, surface area, mean tortuosity, mean angle, and the density of macropores (Perret et al. 2000; Pierret et al. 2002; Bastardie et al. 2003; Peth et al. 2008; Luo et al. 2008; Zhang et al. 2014). Rock fragments are defined as particles >2 mm and can influence the hydrological characteristics of soil by affecting the radius and density of macropores (Flanagan and Nearing 1995; Eriksson and Holmgren 1996; Luo et al. 2010; Shi et al. 2012). Macropores caused by root activity play an important role in the determination of the water flow processes (Stewart et al. 1999; Bundt et al. 2001; Zhang et al. 2016).

The geometry of macropores determines the water infiltration process in soil. However, the quantification of soil macropores remains challenging because of their complicated distribution and geometric features. Traditional methods, such as models for fluid and image analysis of 2D sections by dye tracing, are limited and cannot provide an accurate assessment of the 3D geometry and topology of soil macropores (Liu et al. 2012; Qiu et al. 2013; Katuwal et al. 2015).

In recent years, with technological advances in X-ray CT images, the analysis of the 3D structure of macropores has received increasing attention (Elyeznasni et al. 2012; Hu et al. 2015). Muñoz-Ortega et al. (2015) used CT image analysis and the theory of mathematical morphology to quantify soil macropore characteristics including macropore volume, surface area, connectivity, and size distribution. Dal Ferro et al. (2015) used integrated microscopic CT 3D imaging and pore morphology to characterize total porosity, macropore size distribution, tortuosity, and discrete compactness. Qiao et al. (2015) used focused ion beam-scanning electron microscopy and CT scanning to analyze macropore structure.

Significant differences in the soil macropore distribution occurred upon different land use types (Bottinelli et al. 2016). The soil structures within forest soil may differ in different forest communities. However, studies concerning interactions between forest communities and soil macropores are rare, particularly for the mountainous area in North China. Therefore, the objectives of this study were to (1) use precise industrial CT technology to quantify soil 3D macropore structure and (2) compare and analyze the characteristics and the vertical variation of soil macropores in different forest communities. This study was designed to provide an improved understanding of the connections between forest communities and macropore characteristics in the northern mountainous area in China.

2 Materials and methods

2.1 Experimental site and soil sampling

The experimental site is located in Wuzoulou Forest Station, 85 km northeast of Beijing, China (Fig. 1). This region has a typical semi-humid and semi-dry climate with an annual rainfall of 661 mm and an annual mean temperature of 10.8 °C. The majority of the rock in the study area is derived from Archean granite, and the mountainous area is mainly covered by leptosols according to the typology of the Harmonized World Soil Database (FAO/IIASA/ISRIC/ISS-CAS/JRC 2009.). Vegetation coverage ranged from 40 to 60 %. Plants such as *Platycladus orientalis*, *Quercus* spp., *Pinus tabulaeformis*, *Castanea mollissima*, *Ulmus pumila*, and *Juglans regia* have been cultivated here for 54 years.

Six sites of different forest communities at the same elevation and slope aspect were chosen for this study. The forest community types included (1) mixed *P. tabulaeformis*, *C. mollissima*, and *U. pumila* forest; (2) mixed *P. tabulaeformis* and *C. mollissima* forest; (3) pure forest of *U. pumila*; (4) *C. mollissima*; (5) *J. regia*; and (6) *P. tabulaeformis*. All communities are typical for the local area, and the basic plot is shown in Table 1.

A total of 18 intact soil columns (diameter 100 mm, length 300 mm; three each from the six forest communities) were selected randomly. First, a length of polyvinyl chloride (PVC) pipe was put into the soil surface. Then, a backhoe was used to carefully remove the soil from around the pipe and push the PVC pipe vertically deeper into the soil before later removing the soil column slowly. To avoid disturbing the soil, two iron plates were fixed at the upper and lower ends of each pipe to prevent soil movement and leakage. The iron plates were covered with plastic film to stabilize the soil sample and reduce soil water evaporation. All intact soil columns were immediately transported to the laboratory for CT scanning. In addition, three aluminum boxes of soil (diameter 50 mm, length 50 mm) were taken at three levels (0–10, 10–20, and 20–30 cm) and later used to determine the moisture and nutrient content. Intact soil cores (diameter 100 mm, length 100 mm) were taken at three levels to measure soil particle size distribution and bulk density (Table 2); these soil cores (diameter 70 mm, height 50 mm) were used to measure fine root length density using WinRHIZO (STD4800; Yan et al. 2011).

2.2 CT scanning and image analysis

All intact soil columns were scanned using an industrial cone-beam CT system (Beijing Granpect Co. Ltd., Beijing, China) with a high energy level of 450 keV and 10 mA at a scan speed of 40 mm/rotation. The scanning interval was a continuous 0.215 mm. After scanning and reorientation, 1024 × 1024

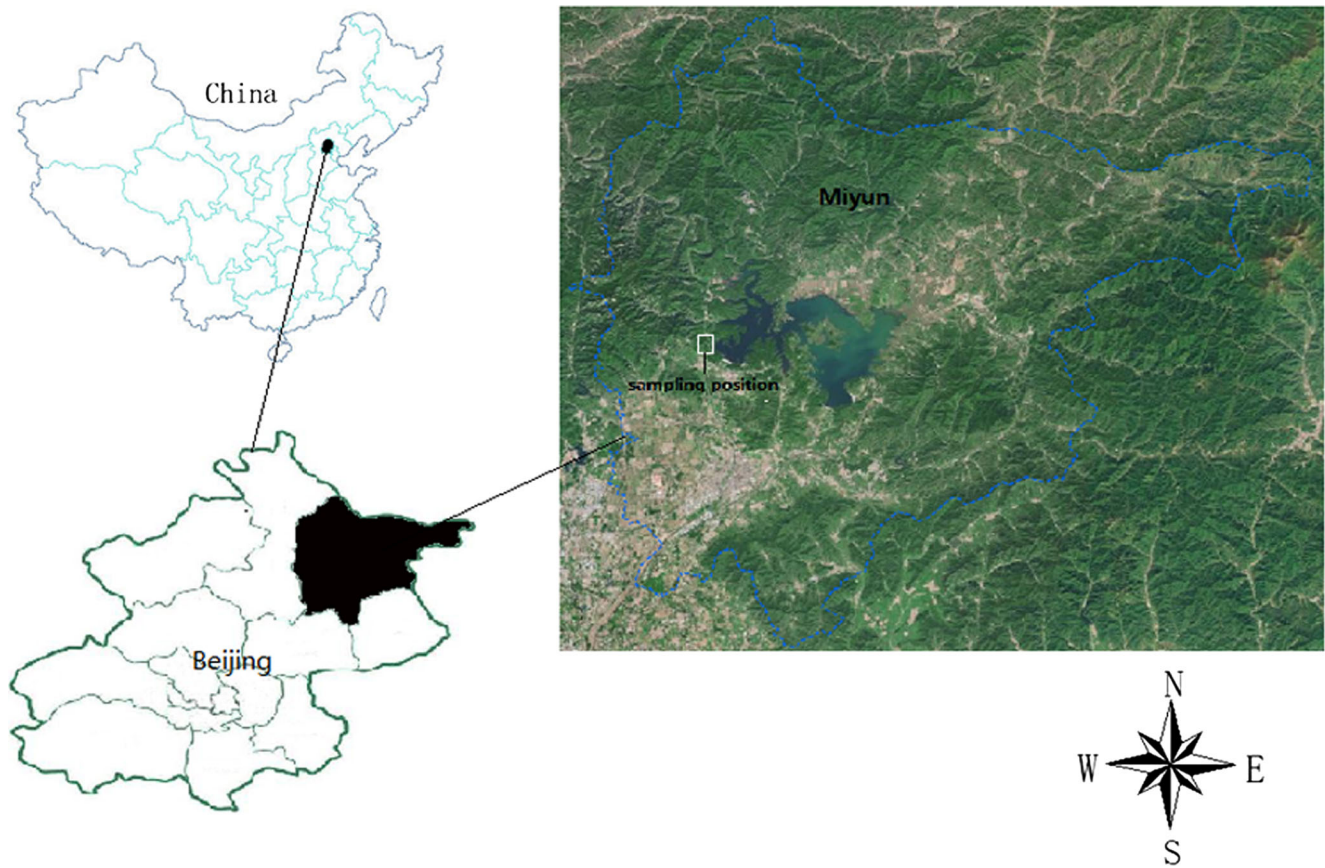


Fig. 1 Study sites and sampling position

images with a voxel size of $0.137 \times 0.137 \times 0.200$ mm were produced.

The image analysis procedure is illustrated in Fig. 2. Commercially available software (VG Studio Max 2.2, Volume Graphics GmbH, Heidelberg, Germany) was used to resample all the CT images and to examine the rock fragment and macropore characteristics. The region of interest (ROI)

tool was used to select the area to be measured and to exclude a 5-mm edge and 35 mm at both ends to avoid the influence of any artificial forces caused by the pipe wall. Values lower than 75 (within the range of 0–255) were selected as the minimum threshold value of macropore size and values greater than 165 were selected as the minimum threshold value of rock fragments. These threshold values were based on observations of

Table 1 Basic situation of sample plots

Forest communities	Coordinate	Study area (m × m)	Altitude (m)	Origin	Forest mean height (m)	Canopy density(%)	Mean DBH (cm)
<i>P. tabuliformis</i> , <i>C. mollissima</i> , and <i>U. pumila</i>	40° 30' 10.81" N, 116° 48' 47.70" E	50 × 50	225	Artificial	11.91	90	22.84
<i>P. tabulaeformis</i> and <i>C. mollissima</i>	40° 30' 09.56" N, 116° 48' 43.95" E	50 × 50	227	Artificial	10.29	90	25.63
<i>U. pumila</i>	40° 30' 10.38" N, 116° 48' 46.29" E	50 × 40	225	Artificial	12.63	85	16.58
<i>C. mollissima</i>	40° 30' 18.30" N, 116° 48' 47.70" E	40 × 40	219	Artificial	8.64	80	28.34
<i>J. regia</i>	40° 30' 19.22" N, 116° 48' 53.71" E	50 × 40	218	Artificial	7.54	80	16.73
<i>P. tabuliformis</i>	40° 30' 19.22" N, 116° 48' 53.71" E	50 × 50	217	Artificial	11.25	85	24.46

DBH diameter at breast height of forest

Table 2 Basic soil properties

Forest communities	Depth (cm)	Bulk density (g cm ⁻³)	Water content (%)	Sand (%)	Silt (%)	Clay (%)	Organic matter (%)	Total potassium (g kg ⁻¹)	Total nitrogen (g kg ⁻¹)	Total phosphorus (mg kg ⁻¹)	Root length density (cm cm ⁻³)
<i>P. tabuliformis</i> , <i>C. mollissima</i> , and <i>U. pumila</i>	0–10	1.58	14.14	65.80	34.20	0.00	5.10	0.73	0.12	5.22	5.28
	10–20	1.52	11.54	67.46	32.54	0.00	4.98	0.18	0.11	4.94	3.80
	20–30	1.54	10.74	58.06	35.80	0.00	2.11	0.08	0.06	4.99	3.31
<i>P. tabulaeformis</i> and <i>C. mollissima</i>	0–10	1.62	10.91	64.20	35.80	0.00	4.84	0.93	0.19	5.01	4.49
	10–20	1.69	16.47	68.75	31.25	0.00	3.61	0.25	0.11	5.69	3.12
	20–30	1.52	10.37	85.52	14.48	0.00	1.35	0.11	0.08	5.15	2.25
<i>U. pumila</i>	0–10	1.66	19.12	74.71	25.07	0.22	2.58	1.44	0.24	7.18	2.13
	10–20	1.41	18.29	66.72	32.88	0.40	2.22	1.03	0.18	5.22	1.69
	20–30	1.74	15.68	67.57	32.20	0.23	0.98	0.52	0.13	5.08	1.58
<i>C. mollissima</i>	0–10	1.56	16.49	88.85	11.15	0.00	1.60	0.18	0.10	6.37	2.60
	10–20	1.60	13.29	93.98	6.02	0.00	1.11	0.11	0.05	4.74	1.55
	20–30	1.47	12.34	91.18	8.82	0.00	0.78	0.11	0.05	5.01	1.11
<i>J. regia</i>	0–10	1.55	12.55	85.58	14.42	0.00	3.41	1.44	0.30	6.03	2.35
	10–20	1.64	11.69	91.17	8.83	0.00	2.20	0.93	0.17	5.35	1.25
	20–30	1.66	13.27	89.56	10.44	0.00	1.74	0.42	0.07	5.49	1.07
<i>P. tabuliformis</i>	0–10	1.49	15.36	50.51	49.49	0.00	1.88	0.52	0.10	4.88	2.48
	10–20	1.69	15.85	65.82	34.18	0.00	1.50	0.21	0.08	5.08	1.12
	20–30	1.76	15.72	50.39	49.30	0.31	1.53	0.14	0.09	7.92	0.81

macropores and rock fragments (Hu et al. 2015, 2016), and all images were inspected visually to ensure the reasonableness of the threshold setting.

After segmentation, 3D rendering and visualization of rock fragments and pore networks were done using VG Studio Max 2.2. Next, we calculated rock fragments and all macropore (macropore = pore diameter > 0.3 mm) characteristics using the Particle Analysis tool of VG Studio Max 2.2 and estimated the volume of rock fragments as well as the diameter, volume, length, number, and distribution of macropores. The surface area of macropores was estimated with the Analyze Particles tool using ImageJ2x.

The angle of a macropore (θ) is defined as the inclination of a macropore branch and could not be obtained directly using the software; therefore, instead, we calculated the angle based on the distribution of macropores. Assuming that all the macropores were not bent, Eq. (1) was used to calculate the angle of each macropore.

$$\theta = \tan^{-1} \frac{z}{\sqrt{x^2 + y^2}} \quad (1)$$

where x , y , and z are the projected macropore lengths in the X , Y , and Z directions, respectively (Fig. 3).

Assuming that all the macropores are cylindrical, the actual macropore length (L_t) was calculated as the macropore volume (V) divided by the macropore surface area (S):

$$L_t = \frac{V}{S} \quad (2)$$

The tortuosity (∞) is used to indicate the degree of convolution of a macropore, which was calculated as L_t divided by the straight line length (L_l):

$$\infty = \frac{L_t}{L_l} \quad (3)$$

The mean tortuosity ($\overline{\infty}$) was calculated using Eq. (4):

$$\overline{\infty} = \frac{\sum_{i=1}^n L_t}{\sum_{i=1}^n L_l} \quad (4)$$

where i is the index of a macropore and n is the total number of macropores (Luo et al. 2010).

2.3 Statistical analyses

One-way analysis of variance (ANOVA) and Fisher's protected least significant difference (LSD) tests were used to evaluate significance using SPSS 22.0 software, at the $p=0.05$ level of confidence.

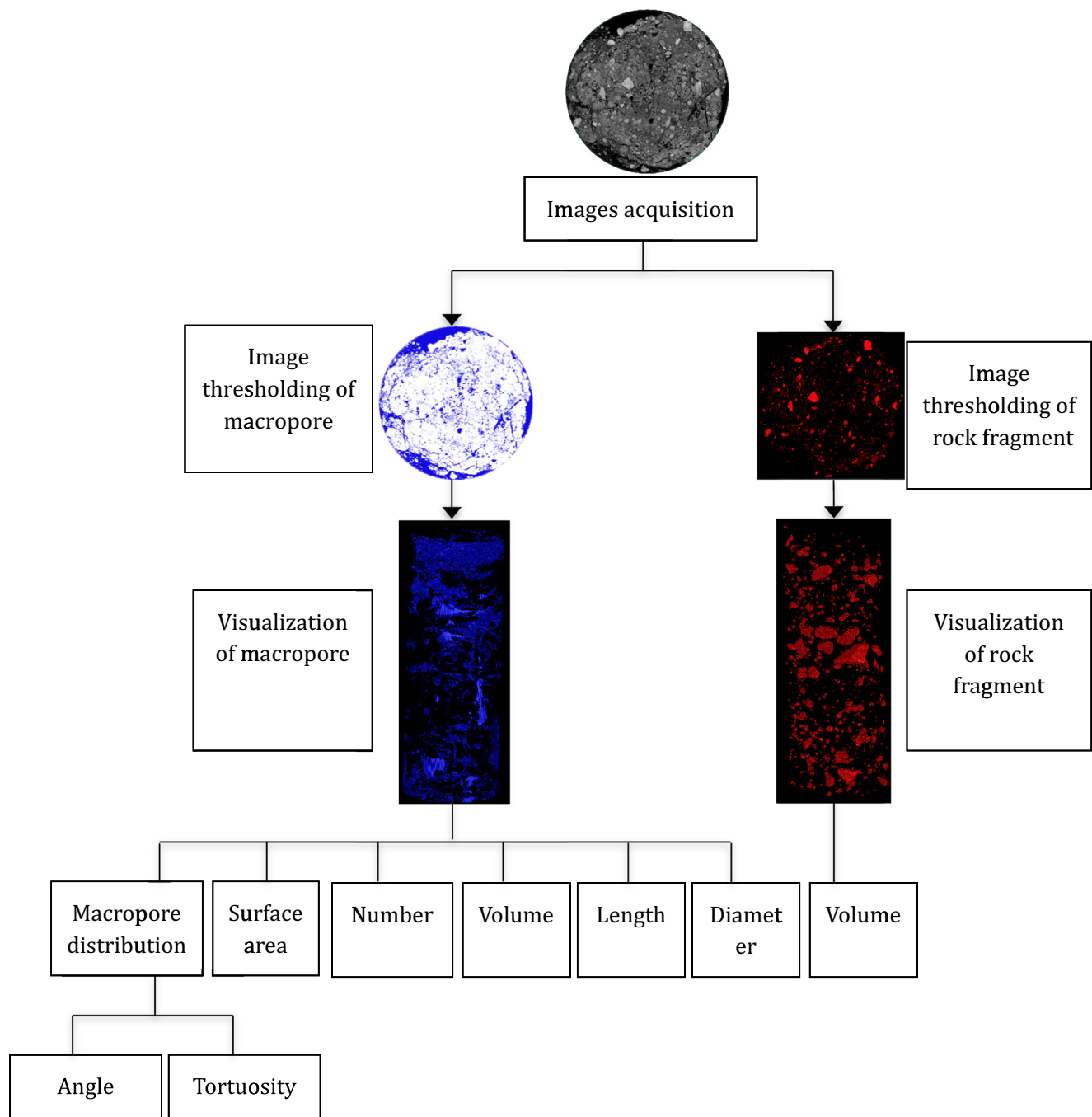


Fig. 2 The image analysis procedures in this study

3 Results

3.1 Visualization of macropore and rock fragment

Figure 4 shows the 3D visualizations of rock fragments in six forest community soils. The distribution of rock fragments was distinct among the different forest communities. The average rock fragment volumes were 5402 ± 1042.22 , 15625 ± 2393 , 12099 ± 2491 , 17208 ± 4516 , 20104 ± 12250 , and

$10872 \pm 3982 \text{ mm}^3$ in the six forest types: mixed *P. tabulaeformis*, *C. mollissima*, and *U. pumila* forest, mixed *P. tabulaeformis* and *C. mollissima* forest, as well as in pure forests of *U. pumila*, *C. mollissima*, *J. regia*, and *P. tabulaeformis*, respectively. The rock fragment volume in the deep soil layer was greater than that in the surface soil layer for all forest communities (Table 3). The activities of roots of plants and trees can redistribute soil rock fragments (Phillips et al. 2005); as a result, mixed forest soils had greater

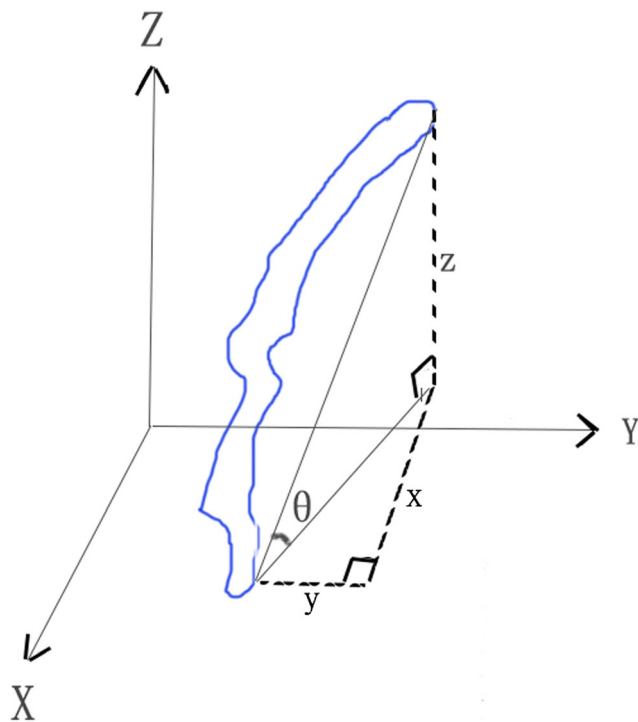


Fig. 3 An example of a macropore 3D distribution

root density than did pure forest soils (Hendriks and Bianchi 1995). However, our results showed that the content of rock fragments in mixed and pure forest soil was not significantly different (Tables 3 and 4); therefore, it is likely that the content of rock fragments in soil at these sites was mainly determined by the parent rock material. Soils under mixed forest communities have greater macroporosity than those of the pure forest communities (Fig. 5). These macropores were both highly continuous and independent. The number of smaller and less continuous macropores in the deep soil layer was greater than that in the surface soil layer under all forest communities. The mixed forest communities had more frequent and regular macropores when compared with the pure forest communities.

3.2 Diameter, volume, surface area, and number of macropores

The total diameter of macropores was significantly different among the six communities (Fig. 6). The total diameter of macropores in soil of mixed *P. tabulaeformis*, *C. mollissima*, and *U. pumila* forest was the highest, slightly higher than that of mixed *P. tabulaeformis* and *C. mollissima* forest, and over twice the diameter density for the pure forest communities analyzed here at all soil depths. Table 3 shows the change of diameter density of macropores for the three soil depth layers in each forest community. All the communities showed a similar pattern of macropore density, in which the diameter density in deep soil was less than that in surface soil.

The total volume, total surface area, and number of macropores varied among the different forest community soils, and the trends were similar to those in the measures of total diameter (Fig. 4). Macroporosity, surface area density, and network density of macropores showed similar vertical variation patterns with diameter density (Table 3).

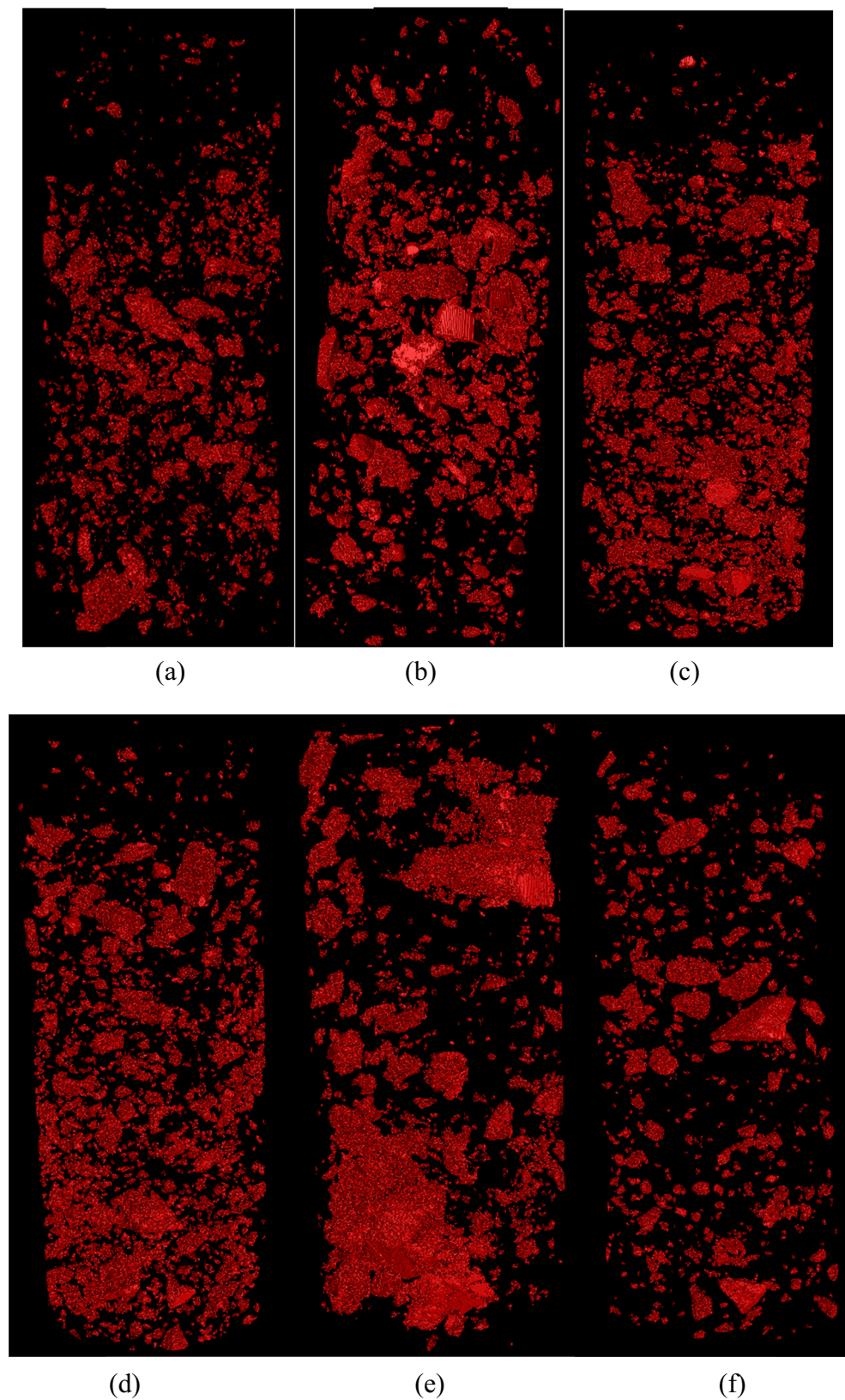
3.3 Tortuosity and angle of macropores

The mean tortuosity of macropores was 1.25 ± 0.13 , 1.29 ± 0.08 , 1.39 ± 0.04 , 1.53 ± 0.10 , 1.61 ± 0.06 , and 1.59 ± 0.01 in soil of six forest types: mixed *P. tabulaeformis*, *C. mollissima*, and *U. pumila* forest, mixed *P. tabulaeformis* and *C. mollissima* forest as well as pure forest of *U. pumila*, *C. mollissima*, *J. regia*, and *P. tabulaeformis*. Figure 6 shows the mean tortuosity of the different soil layers. The mean tortuosity was different among forest communities, and all mixed forest soils showed lower mean tortuosity than soils in pure forest. However, the tortuosity at 20–30 cm soil depth in mixed *P. tabulaeformis*, *C. mollissima*, and *U. pumila* forest was greater than that of mixed *P. tabulaeformis* and *C. mollissima* forest and pure forest of *U. pumila*; in addition, the macroporosity, surface area density, and network density of macropores decreased in the 20–30 cm soil layer in all these forest types, which may be caused by lower root activity, leading to larger tortuosity at that depth. Table 3 shows that the tortuosity values of deep soil layers were higher than those of surface soil layers. No significant differences were found in the mean angle of macropores between soils in different forest communities or at different soil depths (Fig. 4 and Table 3).

3.4 Effect of forest communities on the characteristics of macropores

The analysis of variance indicated that forest community type had significant effects on the length density and network density of macropores in the 0–30 cm soil layers (Table 4). The diameter density of macropores and macroporosity were significantly different between mixed forest soils and pure forest soils in the 0–10 cm soil layer. However, the mean angle of macropores and rock fragment volume showed no significant differences between mixed and pure forest soils. The mean difference values of length density, network density, diameter density of macropores, and macroporosity in the 0–30 cm soil layers between mixed *P. tabulaeformis*, *C. mollissima*, and *U. pumila* forest and mixed *P. tabulaeformis* and *C. mollissima* forest were less than those between mixed *P. tabulaeformis*, *C. mollissima*, and *U. pumila* forest and other pure forest communities analyzed here. The mean difference values of length density, network density,

Fig. 4 3D visualization of soil rock fragment for the soil columns (90 mm in diameter and 230 mm in vertical length after cutting) of mixed forest communities (*P. tabuliformis*, *C. mollissima*, and *U. pumila* (a), *P. tabulaeformis* and *C. mollissima* (b)) and pure forest communities (*U. pumila* (c), *C. mollissima* (d), *J. regia* (e), and *P. tabuliformis* (f)). The red is rock fragment, and the black is not rock fragment



diameter density of macropores, and macroporosity in the 10–30 cm soil layers were significantly less than those in

the 0–10 cm soil layers. Although the difference of surface area density between mixed and pure forest soils was

Table 3 Macropore characteristics of all soil columns

Forest communities	Soil layer (cm)	Rock fragment (%)	Diameter density (mm)	Macroporosity (mm ³ mm ⁻³)	Surface area density (mm ² mm ⁻³)	Network density (number mm ⁻³)	Length density (mm mm ⁻³)	Mean tortuosity	Mean angle (°)
<i>P. tabuliformis</i> , <i>C. mollissima</i> , and <i>U. pumila</i>	0–10	0.1189±0.0700	0.2310±0.0604	0.0319±0.0017	0.0236±0.0020	0.2652±0.0258	0.3125±0.0307	1.1589±0.1514	55.0513±0.6768
	10–20	0.2444±0.0218	0.1687±0.1256	0.0227±0.0131	0.0162±0.0082	0.1942±0.0615	0.2246±0.0768	1.2013±0.1460	54.4309±0.1612
	20–30	0.7474±0.1355	0.1051±0.0549	0.0164±0.0046	0.0116±0.0029	0.1372±0.0373	0.1456±0.0363	1.4060±0.1564	53.7628±0.2371
<i>P. tabulaeformis</i> and <i>C. mollissima</i>	0–10	0.5291±0.1326	0.1944±0.0845	0.0275±0.0080	0.0226±0.0103	0.2337±0.1093	0.2501±0.1407	1.2220±0.1148	49.3292±0.4917
	10–20	1.4454±0.0800	0.1492±0.0521	0.0232±0.0078	0.0179±0.0071	0.1752±0.0495	0.1834±0.0617	1.2718±0.1180	48.9297±0.3887
<i>U. pumila</i>	20–30	1.2378±0.4222	0.1273±0.0537	0.0224±0.0110	0.0168±0.0092	0.1465±0.0307	0.1492±0.0066	1.3948±0.1288	48.7633±0.8403
	0–10	0.5592±0.1153	0.1018±0.0147	0.0197±0.0031	0.0153±0.0027	0.1095±0.0363	0.1208±0.0174	1.3605±0.1710	56.4314±1.5886
	10–20	0.7396±0.1174	0.0985±0.0227	0.0209±0.0026	0.0148±0.0028	0.0912±0.0314	0.1181±0.0235	1.4329±0.0276	56.5039±1.2659
<i>C. mollissima</i>	20–30	1.1886±0.3957	0.0524±0.0133	0.0115±0.0056	0.0087±0.0048	0.0560±0.0294	0.0637±0.0171	1.4060±0.1145	55.6625±1.9372
	0–10	0.9307±0.1512	0.1112±0.0101	0.0128±0.0043	0.0159±0.0032	0.1563±0.0159	0.0847±0.0084	1.4827±0.1586	51.5882±0.0881
	10–20	0.9439±0.3953	0.1022±0.0161	0.0108±0.0038	0.0135±0.0019	0.1431±0.0006	0.0787±0.0099	1.4589±0.2347	51.5990±0.6020
<i>J. regia</i>	20–30	1.6632±0.8326	0.0756±0.0203	0.0083±0.0032	0.0099±0.0024	0.1095±0.0269	0.0581±0.0167	1.6482±0.0866	51.7658±0.1797
	0–10	1.1840±0.7149	0.0672±0.0238	0.0131±0.0053	0.0116±0.0040	0.0801±0.0268	0.0549±0.0190	1.6307±0.2011	49.2637±0.5846
	10–20	0.9589±0.2531	0.0622±0.0209	0.0118±0.0045	0.0112±0.0056	0.0755±0.0219	0.0515±0.0159	1.6627±0.2577	49.3931±0.4214
<i>P. tabuliformis</i>	20–30	1.9902±1.7046	0.0337±0.0137	0.0057±0.0013	0.0057±0.0028	0.0387±0.0190	0.0269±0.0119	1.5413±0.2425	48.8327±0.5728
	0–10	0.6298±0.1574	0.1114±0.0275	0.0101±0.0032	0.0113±0.0023	0.1485±0.0402	0.0799±0.0248	1.6065±0.0735	50.8055±1.7477
	10–20	0.7312±0.4090	0.0937±0.0200	0.0082±0.0045	0.0095±0.0045	0.1297±0.0284	0.0675±0.0164	1.5826±0.0710	50.8955±2.0806
	20–30	0.8742±0.3135	0.0294±0.0348	0.0019±0.0019	0.0027±0.0026	0.0422±0.0448	0.0235±0.0189	1.5969±0.1536	49.9265±4.7178

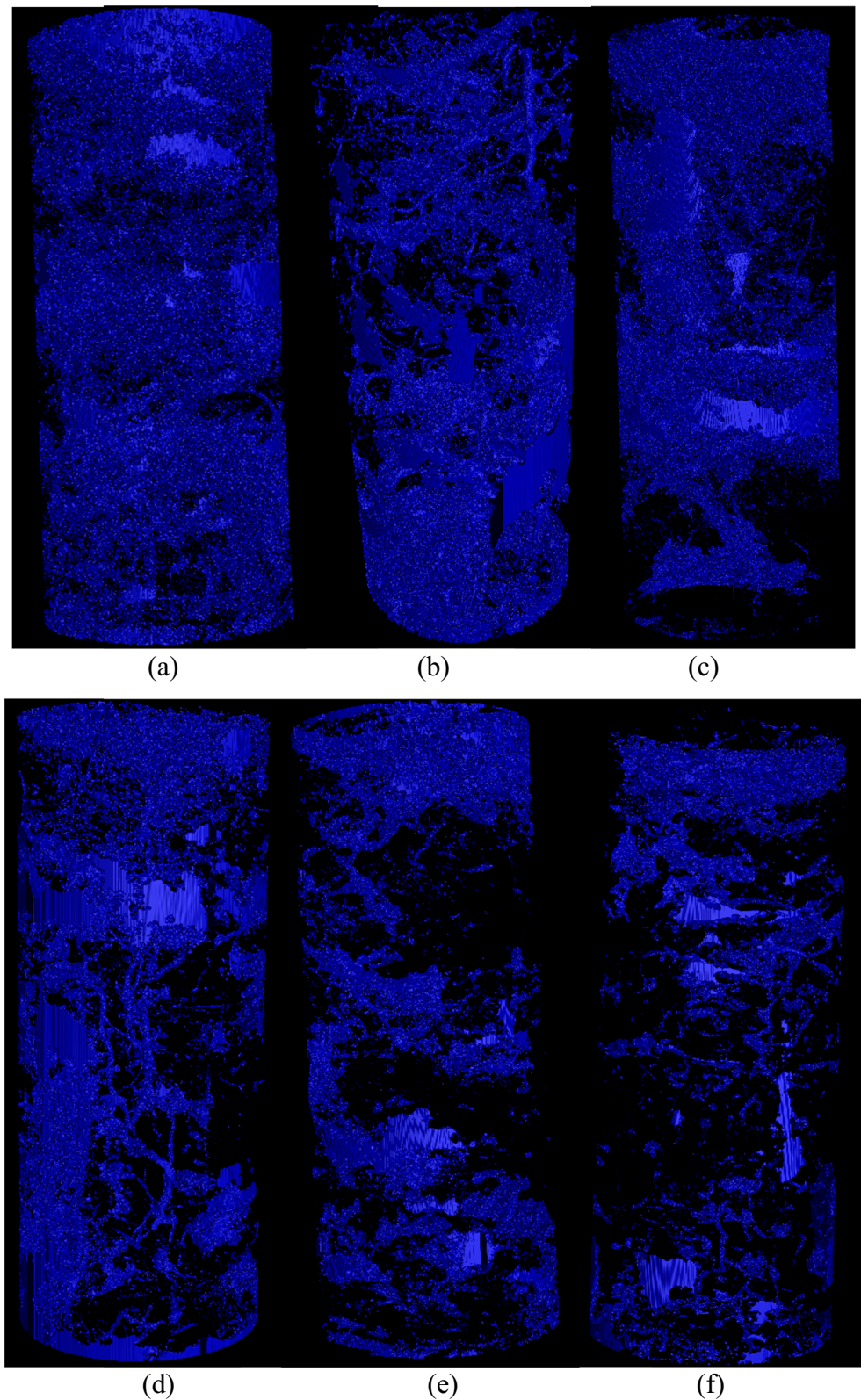
Rock fragment (>2 mm in size) volume was calculated by CT images analyzed. Length density is described using macropore straight line length. Diameter density was calculated from total diameter divided by column total volume. Surface area density was determined by total surface area divided by column total volume. Length density was calculated from total length of macropores divided by column total volume

Table 4 Analysis of variance of macropore characteristics of different forest community soils at three soil layers

Soil layer (cm)	Communities	Length density (mm)	Network density (number mm ⁻³)	Diameter density (mm mm ⁻³)	Macroporosity (mm ³ mm ⁻³)	Surface area density (mm ² mm ⁻³)	Mean angle (°)	Mean tortuosity	Rock fragment volume (%)
0–10	a	-0.0625 (0.607a)	-0.0315 (0.476a)	-0.0366 (0.344a)	-0.0045 (0.670a)	-0.0009 (0.812a)	-5.7221 (0.000b)	0.0630 (0.618a)	0.0041 (0.136a)
	b	-0.1918 (0.009b)	-0.1557 (0.003b)	-0.1292 (0.005b)	-0.0122 (0.031b)	-0.0083 (0.065a)	1.3801 (0.134a)	0.2015 (0.128a)	0.0044 (0.112a)
	c	-0.2279 (0.003b)	-0.1089 (0.026b)	-0.1198 (0.007b)	-0.0191 (0.001c)	-0.0077 (0.084a)	-3.4631 (0.002c)	0.3238 (0.022b)	0.0081 (0.008b)
	d	-0.2576 (0.001b)	-0.1851 (0.001b)	-0.1638 (0.001b)	-0.0189 (0.002c)	-0.0120 (0.013b)	-5.7876 (0.000b)	0.4717 (0.002b)	0.0107 (0.001b)
	e	-0.2327 (0.002b)	-0.1167 (0.018b)	-0.1196 (0.007b)	-0.0219 (0.000c)	-0.0122 (0.011b)	-4.2458 (0.000b)	0.4475 (0.003b)	0.0051 (0.070a)
10–20	a	-0.0412 (0.657a)	-0.0190 (0.548a)	-0.0195 (0.687a)	0.0005 (0.927a)	0.0017 (0.711a)	-5.5012 (0.000b)	0.0705 (0.610a)	0.0120 (0.000b)
	b	-0.1065 (0.038b)	-0.1030 (0.006b)	-0.0702 (0.164a)	-0.0018 (0.765a)	-0.0013 (0.771a)	2.0730 (0.029c)	0.2316 (0.111a)	0.0050 (0.038c)
	c	-0.1460 (0.005b)	-0.0511 (0.122a)	-0.0665 (0.185a)	-0.0118 (0.061a)	-0.0026 (0.567a)	-2.8318 (0.005d)	0.2576 (0.079a)	0.0070 (0.007c)
	d	-0.1731 (0.001b)	-0.1187 (0.002b)	-0.1065 (0.044b)	-0.0109 (0.081a)	-0.0050 (0.287a)	-5.0378 (0.000b)	0.4613 (0.005b)	0.0072 (0.006c)
	e	-0.1571 (0.003b)	-0.0645 (0.058b)	-0.0750 (0.139a)	-0.0145 (0.026b)	-0.0067 (0.162a)	-3.5353 (0.001d)	0.3812 (0.015b)	0.0049 (0.041c)
20–30	a	0.0036 (0.831a)	0.0092 (0.734a)	0.0223 (0.467a)	0.0060 (0.214a)	0.0051 (0.211a)	-4.9995 (0.014b)	-0.0112 (0.931a)	0.0049 (0.479a)
	b	-0.0818 (0.001b)	-0.0814 (0.010b)	-0.0523 (0.101b)	-0.0049 (0.309b)	-0.0029 (0.464a)	1.8996 (0.295a)	0.0001 (1.000a)	0.0044 (0.523a)
	c	-0.0874 (0.001c)	-0.0278 (0.314a)	-0.0295 (0.339a)	-0.0081 (0.101b)	-0.0017 (0.665a)	-2.0060 (0.271a)	0.2422 (0.080a)	0.0092 (0.197a)
	d	-0.1186 (0.000d)	-0.0987 (0.003b)	-0.0713 (0.033c)	-0.0107 (0.037b)	-0.0059 (0.155b)	-4.9302 (0.015b)	0.1353 (0.306a)	0.0124 (0.089a)
	e	-0.1221 (0.000d)	-0.0952 (0.004b)	-0.0756 (0.025c)	-0.0144 (0.008b)	-0.0090 (0.400b)	-3.8364 (0.047b)	0.1909 (0.157a)	0.0013 (0.853a)

The data of mean difference were analyzed using Dunnett's test, mixed forest of *P. tabulaeformis*, *C. mollissima*, and *U. pumila* as the control item. Fisher's LSD test was used to analyze significant difference of six forest communities; the values in brackets are significant level; different letters in the same soil layers means significant correlation ($P < 0.05$). a = mixed forest of *P. tabulaeformis*, *C. mollissima*, and *U. pumila*; b = mixed forest of *P. tabulaeformis* and *C. mollissima*; c = pure forest of *U. pumila*; d = pure forest of *C. mollissima*; e = pure forest of *J. regia*; f = pure forest of *P. tabulaeformis*

Fig. 5 3D visualization of soil macropore structures for the soil columns (90 mm in diameter and 230 mm in vertical length after cutting) of mixed forest communities (*P. tabuliformis*, *C. mollissima*, and *U. pumila* (a) and *P. tabulaeformis* and *C. mollissima* (b)) and pure forest communities (*U. pumila* (c), *C. mollissima* (d), *J. regia* (e), and *P. tabuliformis* (f)). The blue is pore, and the black is non-pore



not significant, the difference of surface area density between mixed *P. tabulaeformis*, *C. mollissima*, and *U. pumila* forest and mixed *P. tabulaeformis* and

C. mollissima forest was smaller than that of mixed *P. tabulaeformis*, *C. mollissima*, and *U. pumila* forest and other pure forest soils.

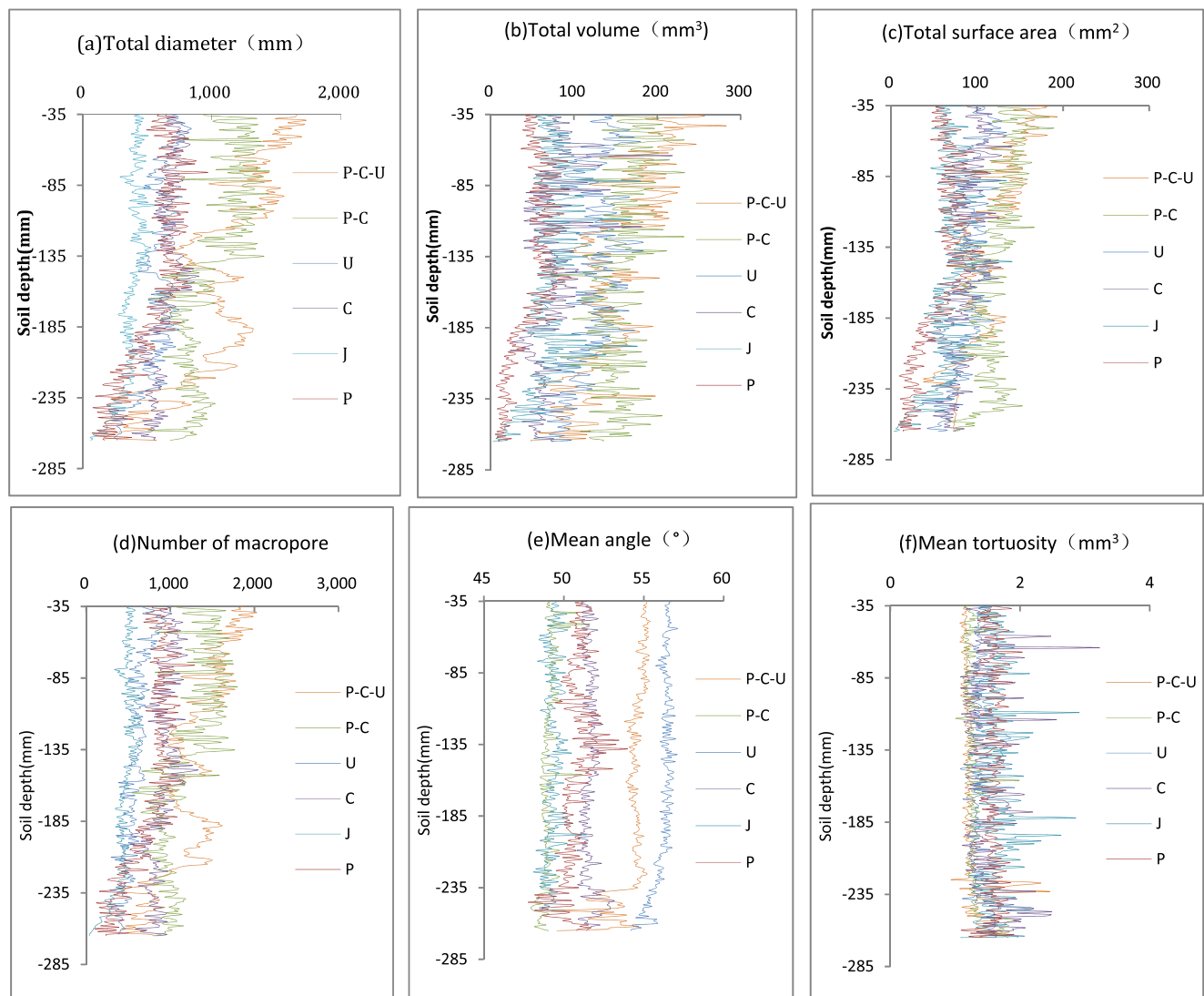


Fig. 6 Total diameter (a), total volume (b), total surface area (c), number (d), mean angle (e), and mean tortuosity (f) of macropores varied among the soil depth in six forest community soils. P-C-U is mixed *P.*

tabuliformis, *C. mollissima*, and *U. pumila* forest; P-C is mixed *P. tabuliformis* and *C. mollissima* forest; U, C, J, and P is pure forest of *U. pumila*, *C. mollissima*, *J. regia*, and *P. tabulaeformis*, respectively

4 Discussion

Industrial CT allowed the accurate calculation of the size and type of macropore structures. Rock fragment volume and some characteristics of macropores, such as straight line length, volume, surface area, diameter, number, and the distribution of macropores can be directly calculated. The actual length of macropores, mean angle, and mean tortuosity were calculated by applying the determined macropore distribution to formulas. However, the actual length of macropores was estimated by assuming that all macropores are cylinders, which may generate some error in the calculation of macropore tortuosity. The development of an improved method for macropore tortuosity calculation will be a focus of future studies. We found that the threshold value during image

analysis had a large effect on rock fragment measurement results, even with very small changes. Because human ocular observation was used to set the threshold values, the thresholds of large density soil and rock fragments were very similar, and the rock fragment volume might be inaccurately estimated. A more accurate method for distinguishing the threshold for rock fragments would allow a more accurate analysis.

Hendriks and Bianchi (1995) reported that root density decreased gradually with soil depth in both mixed and pure forests. Our results demonstrated that the size of macropores (volume, diameter, surface area) also decreased with soil depth in soils of forest communities, because of the greater numbers of roots and increased biota activity in the soil surface, consistent with the findings of Luo et al. (2010). In addition, macropores in

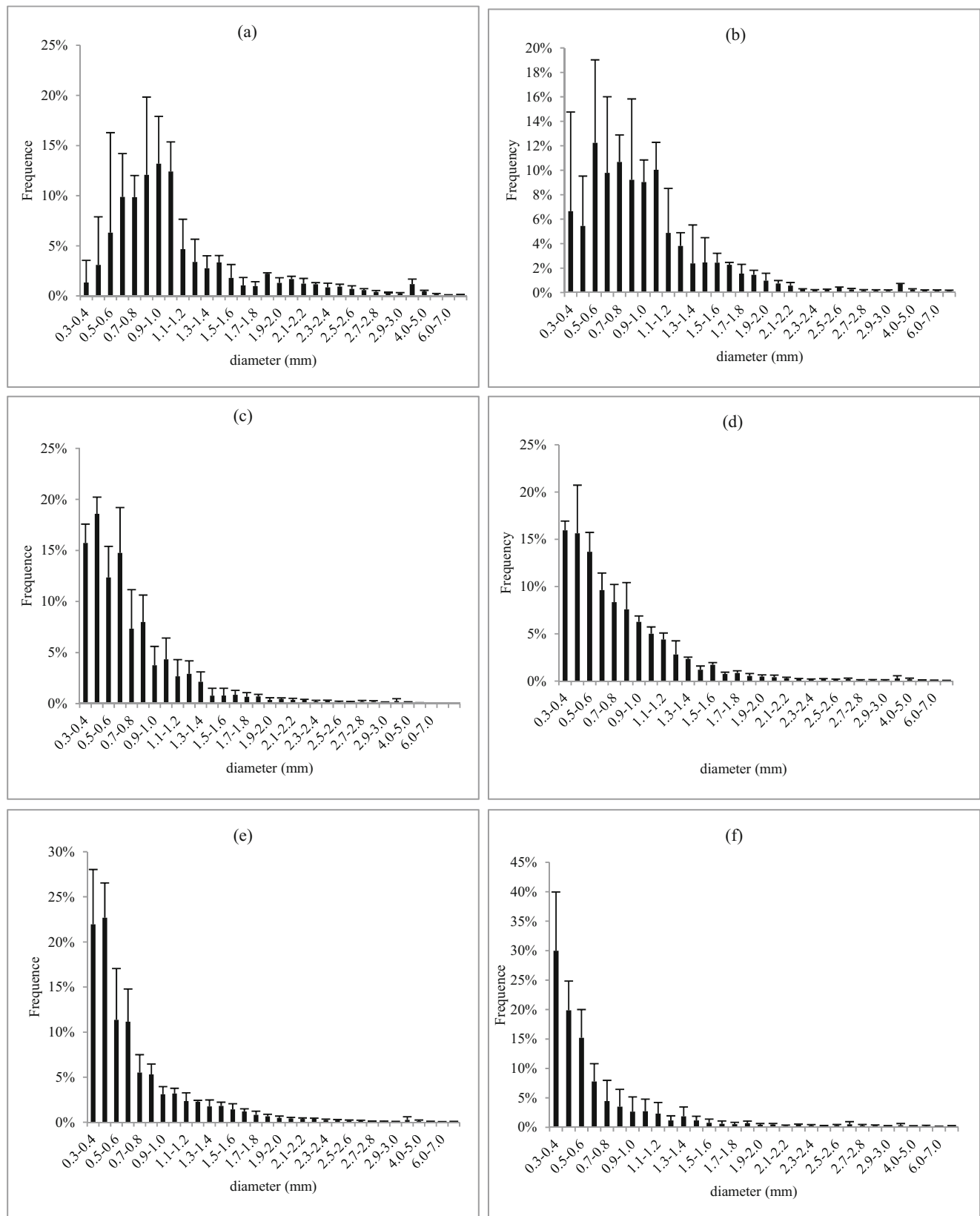


Fig. 7 Diameter distribution of macropores in mixed forest communities (*P. tabuliformis*, *C. mollissima*, and *U. pumila* (a), *P. tabulaeformis* and *C. mollissima* (b)) and pure forest communities (*U. pumila* (c), *C. mollissima* (d), *J. regia* (e), and *P. tabuliformis* (f))

deeper soil layers showed a higher level of tortuosity, and higher rock fragment volumes were found in deeper soil layers. Soils under mixed forest communities developed deeper macropores when compared with pure forest soils. The macropores in mixed forest soils were distributed at depths of 0–30 cm. In contrast, macropores under pure forest were mostly concentrated at depths of 0–20 cm. The root density in mixed forest soils decreased more gradually with depth than did the density in pure forest soils (Hendriks and Bianchi 1995). One possible reason for this finding is that better root development occurred at deep soil depths in mixed forest, and this resulted in a great number of macropores.

The results of this study suggest that the characteristics of macropores differ between soils of different forest communities especially at a depth of 0–20 cm. Soils of pure forest communities had fewer but more curved macropores when compared with soils of two mixed forest communities; this was caused by the latter's higher activity of biota and greater numbers of roots. Macropores with diameters of 0.4–1.1 mm accounted for 66.83 ± 4.77 % of all macropores in soil of mixed *P. tabulaeformis*, *C. mollissima*, and *U. pumila* forest and 66.47 ± 6.19 % of all macropores in soils of mixed *P. tabulaeformis* and *C. mollissima* forest. In addition, macropores with diameters of 0.3–0.7 mm accounted for 64.09 ± 10.11 % of all macropores in soils of all pure forest communities studied here (Fig. 7). The soil macropores in the mixed forest were larger than those in pure forest, which will allow more water to infiltrate into the soil. However, soils with higher soil organic matter content in mixed forest communities generally helps to increase the stability of the soil structure (Table 2). Root length density in mixed forest soil is higher than that in pure forest soil (Table 2), and mixed forest soil had a greater density and smaller tortuosity of macropores than did pure forest soil (Table 3). These results are consistent with previous studies, which found that the presence of roots resulted in increased macropore density and reduced macropore tortuosity (Li et al. 2009; Hu et al. 2015) and that the root density in mixed forest soils was higher than that in pure forest soils (Hendriks and Bianchi 1995).

Zhang et al. (2016) reported that macropores have a great effect on water flow although they comprise only a small fraction of total porosity. Large amounts of macropores increase the infiltration rate and reduce surface runoff (Yang and Zhang 2011). Saravanathiiban et al. (2014) reported that the preferential flow rate along macropores decreased in areas with high tortuosity, and our results show that mixed forest soils have lower tortuosity than pure forest soils (Table 3). Therefore, the results of this study also imply that mixed forest soils have a more rapid preferential flow rate when compared with pure forest soils. We suggest that the soil macropore architecture of mixed forest could increase water transport

to deeper soil layers when compared with that in pure forest, which could reduce surface runoff and decrease erosion.

5 Conclusions

The analysis of the characteristics of macropores in forest community soil using industrial CT and graphic analysis is shown here to be a promising method. A variety of parameters related to macropores were quantified, and the finding demonstrated the effects of various forest communities on the nature and density of macropores. Soils under a mixed forest community have more macropores than soils in a pure forest community. There are more macropores in the deep soil of a mixed forest than in that of a pure forest. Moreover, macropores in soils of mixed forest communities are more vertical and convoluted than those in soils of pure forest communities.

Acknowledgments The National Science Foundation of China (NSFC 41271044) and the Graduate Training and Development Program of Beijing Municipal Commission of Education partially supported this study.

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